

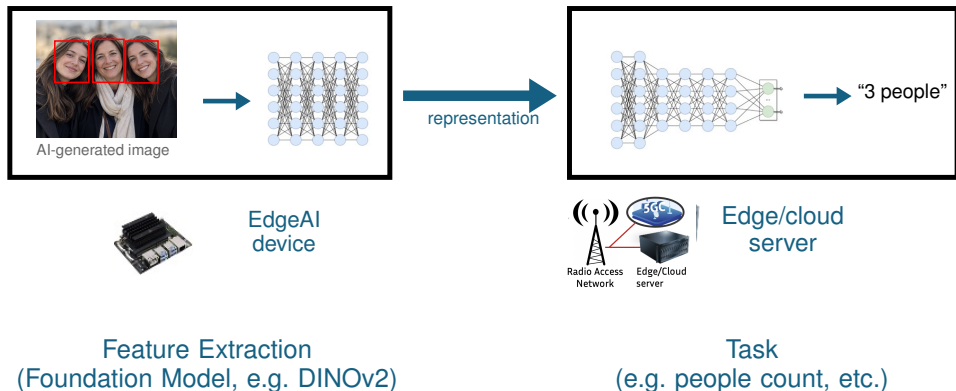
Privacy of Representations in the Era of Transformer-based Vision Foundation Models

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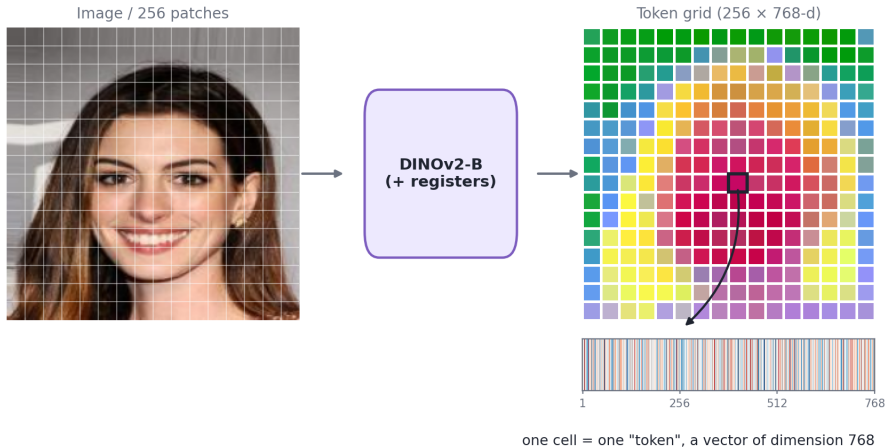
How much privacy is leaking in Vision Foundation Models?



**Foundation models are shared backbones;
in split computing the server receives their features.**

The server is **honest-but-curious**: it follows the protocol, but tries to infer private information.

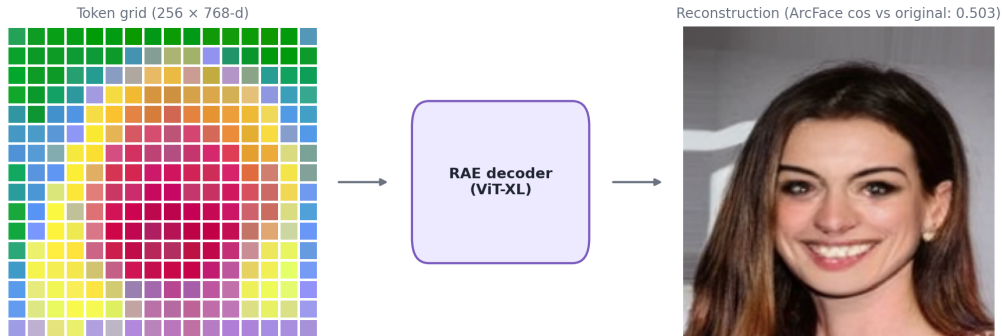
What does the server actually see?



The server receives the full token grid, not just one pooled vector (CLS token).

Vision Transformer on device: **DINOv2 with registers** $\Rightarrow$ 256 × 768 token grid

The token grid can be inverted back to an image



Input inversion is no longer hard for the latent space of DINOv2-type models with RAE^[1].

^[1]RAE: Zheng et al., ICLR 2026

State of the art

- Privacy of latent spaces: many works on CNNs, **few on Vision Transformers**
- Many proposed protections did not withstand adaptive attacks
- Diffusion / flow-matching models are strong **priors**: they know what natural images look like, so they help the attacker rebuild the image (DRAG^[1])

The open question

FMTM^[2] shows that a **linear projection**, “Identity Sanitization Projection” (**ISP**), protects the compact CLS vector of foundation models. What about applying ISP to the **full token grid** that split inference actually transmits?

^[1]DRAG: Lei et al., PMLR 2025

^[2]FMTM: George et al., 2026

Our contribution

① How much privacy leaks from the full token grid?

- measured with face recognition (ArcFace), Type-II SAR @ FAR= 10^{-4}

② With which attacks?

- a distilled model (Token2ArcFace) and RAE reconstruction
- attacks:
 - **naive**: trained on undefended tokens
 - **adaptive**: re-trained on defended tokens

③ How effective are the protections on the full token grid?

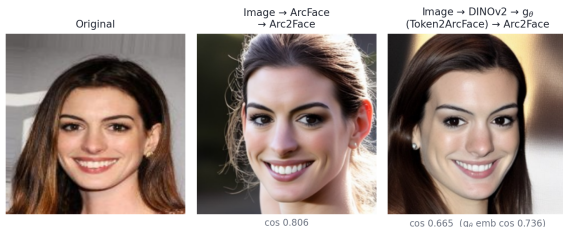
- linear projection (ISP) vs. gradient-based noise

④ How much utility is preserved when the server runs its task?

- face segmentation quality on the defended tokens, scored by mIoU

How do we measure privacy leakage?

- Pixel error (MSE) is a poor privacy measure: an attacker only needs the **identity embedding**
- Face recognition (ArcFace) turns a face into a **512-number identity embedding**
- Two faces match if their embeddings have a high **cosine similarity**
- From it, a generative model (Arc2Face^[1]) can **regenerate the face**, even from a “protected” input



^[1]Arc2Face: Papantoniou et al., ECCV 2024

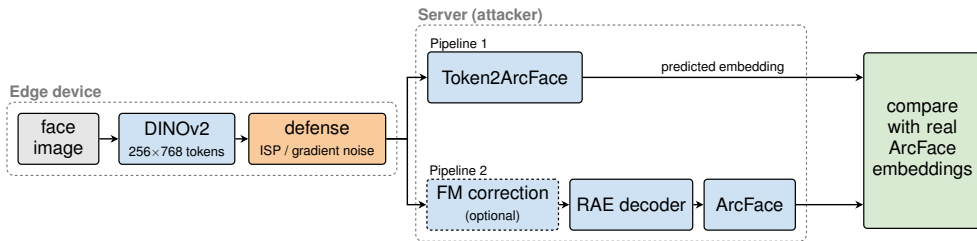
Two defenses, two attack pipelines

Defenses

- **ISP:** project out the r most identity-informative directions in the latent space
- **Gradient-based noise:** noise along directions that most change the ArcFace prediction

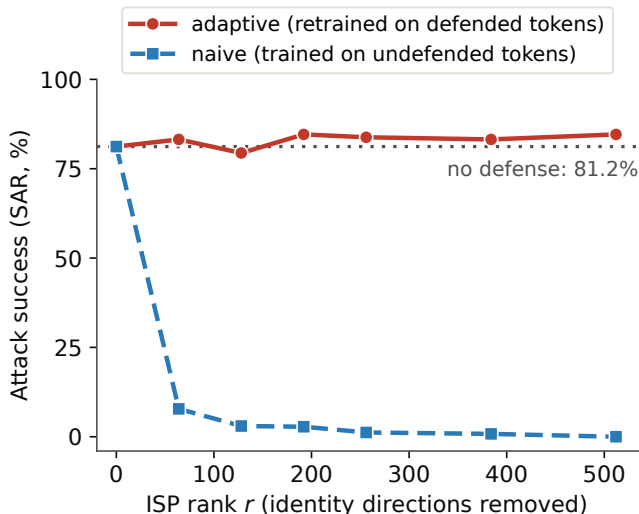
Attacks

- **P1:** Token2ArcFace, distilled from ArcFace (token grid $\rightarrow$ identity embedding)
- **P2:** RAE (+ flow matching) $\rightarrow$ image $\rightarrow$ ArcFace



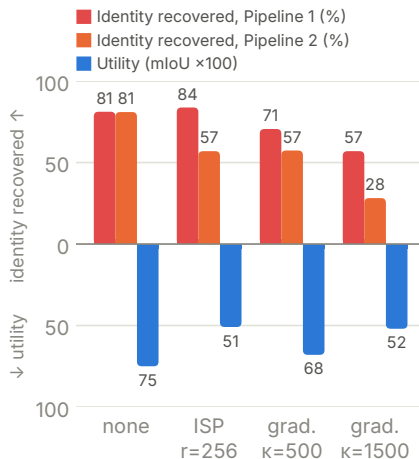
Result 1: ISP is safe only against a naive attacker

- Without any defense, Token2ArcFace recognises the face **81.2%** of the time
- Naive** attacker: drops from 7.8% to **0%**, so ISP looks safe
- Adaptive** attacker: stays at **79.4–84.6%**, so no real protection



Result 2: gradient-based noise still protects, with utility comparable to ISP

- Gradient-based noise **still protects identity**, even against the **adaptive** attacker: across the noise sweep it falls from **70.6%** to **47.4%**, where it saturates
- Utility (segmentation mIoU): **0.75** undefended; ISP keeps **0.51**, but does not protect against the adaptive attacker
- At the **same utility**, gradient-based noise keeps identity leakage far lower than ISP



The graph shows 4 operating points out of the full sweep

Conclusion and future work

Conclusion

- We provided results on **evaluating and improving the privacy of the latent space** of a popular vision model
- The full token grid leaks identity, ISP does not resist an adaptive attacker, while **gradient-based noise still protects**, with comparable utility

Future work

- Further exploiting **FM/diffusion models** for improving utility/privacy tradeoffs (iterating FM and protection steps)
- Improved latent spaces (**RAEv2**)
- **Mechanistic interpretability** methods, to explore better elimination of privacy-sensitive features

Thank you

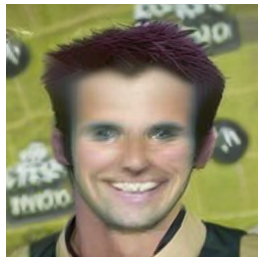
Thank you!

Questions?

Backup: what each defense does to the image



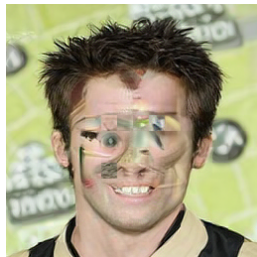
real



ISP $r=256$



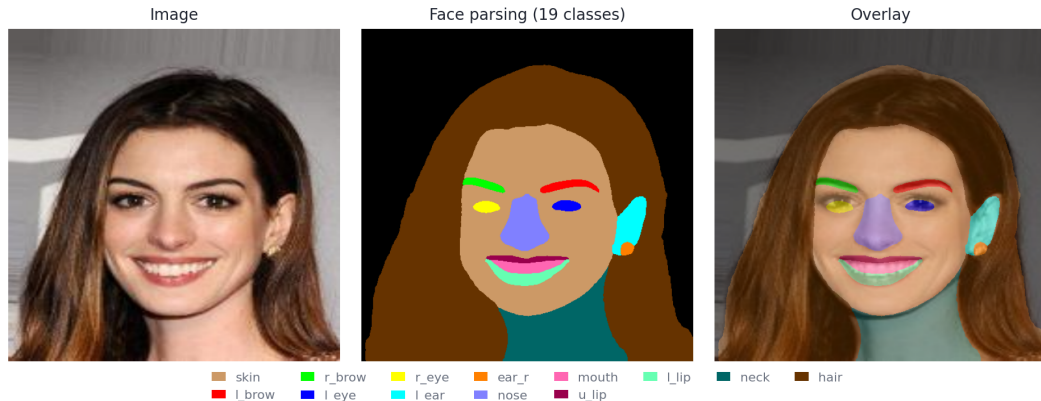
$\kappa=4500$



$\kappa=4500+FM$

ISP removes r identity directions, gradient-based noise has strength κ , and FM is the flow-matching correction applied to the defended tokens before decoding.

Backup: the utility task



The server's own task runs on **the same token grid**: a frozen convolutional head predicts a 19-class face parsing (CelebAMask-HQ), scored by **mIoU**. This is the utility a defense has to preserve.

Backup: the metric, Type-II SAR @ FAR = 10^{-4}

- **What it measures:** how often the attacker's output is accepted as the right person, when the system is calibrated to wrongly accept a *stranger* only **1 time in 10,000**
- **SAR** (successful acceptance rate): the attacker's embedding is accepted if its **cosine similarity** to a reference embedding passes a threshold
- **FAR** (false acceptance rate): that threshold is tuned on the validation split so that impostors pass only 10^{-4} of the time
- **Type-II:** the attacker's output is compared with *other* images of the same person, not with the image it was derived from
- Identity-disjoint train / validation / test sets (6622 / 500 / 500 identities); ISP fitted on a separate 1000 identities